1 Planning (15 marks)

Defining the problem (3 marks)

P1 n is the independent variable and V is the dependent variable or vary n and measure V [1]
P2 Keep distance from light to photocell constant [1]
P3 Keep intensity of light constant. Allow constant voltage across lamp/current through lamp/brightness. Do not allow 'same lamp/output'. [1]

Methods of data collection (5 marks)

M1 Labelled diagram of apparatus: lamp, glass sheet and photocell in line. [1]
M2 Voltmeter connected to photocell. Penalise unworkable photocell circuit. [1]
M3 Use micrometer (screw gauge) to measure thickness of glass sheet. [1]
M4 Take many readings of thickness and average. [1]
M5 Perform experiment in a dark room or shield apparatus. [1]

Method of analysis (2 marks)

A1 Plot a graph of ln V against n. Allow ln V against nt [1]
A2 α = (–)gradient / t. (ln V against nt then α = (–)gradient) [1]

Safety considerations (1 mark)

S Reasoned method to prevent burns from hot source, e.g. use gloves
Reasoned method to prevent eye damage from bright/intense source, e.g. shield lamp/ dark glasses/do not look at source directly
Reasoned method to prevent cuts from glass e.g. use gloves. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use small distance/high intensity to gain large reading.
2 Method to check output of lamp is constant e.g. measure current through/p.d. across lamp/regularly check V_0 with no glass.
3 Reasoned method to ensure output of lamp is constant e.g. workable circuit diagram with variable resistor or variable power supply.
4 Clean sheets of glass before use.
5 Direction of light is perpendicular to glass sheets/constant orientation.
6 $\ln V = -\alpha nt + \ln V_0$.
7 Further safety consideration.

Do not allow vague computer methods.

[Total: 15]

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